Semi-Supervised Landmark Tracking in Echocardiography Video via Spatial–Temporal Co-Training and Perception-Aware Attention

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***Abstract*— Precise landmark annotation in cardiac ultrasound images is fundamental for quantitative cardiac health assessment. However, the time-intensive nature of manual annotation typically constrains clinicians to annotate only selected key frames, limiting comprehensive temporal analysis capabilities. While recent automated landmark detection methods have demonstrated success for key-frame analysis, they fail to effectively utilize the intrinsic temporal information across cardiac sequence. To bridge this gap, we present SemiEchoTracker, a novel semi-supervised framework that enables comprehensive landmark tracking throughout echocardiography sequences while requiring supervision only on key frames. Our framework introduces three key innovative strategies: 1) a co-training mechanism that enforces mutual consistency between spatial detection and temporal tracking, enabling accurate intermediate frame detection without additional annotations, 2) a guided DINOv2 pretraining strategy that is specially tailored for extracting fine-grained echocardiography-specific spatial features, and 3) a perception-aware spatial-temporal (PAST) attention module that efficiently captures inter- and intra-frame relationships in echocardiography videos. Extensive validation on three datasets across multiple cardiac views demonstrates that our method not only achieves state-of-the-art detection performance on the keyframes but also yields accurate frame-by-frame prediction, which is important for dynamic cardiac analysis in clinicians.**

***Index Terms*— Semi-supervised learning, landmark detection and tracking, key frames supervision, echocardiography, spatial–temporal attention mechanism.**

Received 29 August 2025; revised 25 November 2025; accepted 29 December 2025. Date of publication 6 January 2026; date of current version 4 May 2026. This work was supported in part by the National Natural Science Foundation of China under Grant U23A20295, Grant 82441023, Grant 62131015, and Grant 82394432; in part by China Ministry of Science and Technology under Grant S20240085, Grant STI2030-Major Projects-2022ZD0209000, and Grant STI2030-Major Projects-2022ZD0213100); in part by Shanghai Municipal Central Guided Local Science and Technology Development Fund under Grant YDZX20233100001001; in part by the Key Research and Development Program of Guangdong Province, China, under Grant 2023B0303040001; and in part by HPC Platform of ShanghaiTech University. Recommended by Associate Editor W.-N. Lee. *(Corresponding authors: Lin Zhou; Qi Xu; Zhiming Cui; Dinggang Shen.)*

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Digital Object Identifier 10.1109/TMI.2026.3651389

I. Introduction

ECHOCARDIOGRAPHY serves as the cornerstone of cardiac function assessment [1], playing a vital role in cardiovascular disease diagnosis [2]. Through precise anatomical landmark placement on echocardiographic images, clinicians can extract comprehensive information about cardiac structures and dynamics, enabling detailed functional assessment and pathological condition identification [3]. However, the manual annotation process requires substantial expertise and time investment, compelling clinicians to restrict their analysis to key frames within existing clinical workflows. This key-frame-only approach, while practical, limits assessment to only static cardiac parameters, potentially overlooking critical temporal dynamics and subtle motion patterns that could provide valuable diagnostic insights [4].

Deep learning has revolutionized echocardiographic landmark analysis, with contemporary approaches evolving along two distinct methodological paradigms: frame-based and video-based methods. Frame-based approaches [5], [6] primarily focus on static landmark detection within individual key frames. While effective for single-frame analysis, their predictions fail to maintain temporal consistency across frames, inherently limiting their capability for comprehensive temporal analysis of cardiac motion. Video-based methods [6], [7] have

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advanced to incorporate temporal information through the utilization of the whole sequence as input. However, their supervision remains confined to key frames with no supervision for analysis of the intermediate frames, which constrains their ability to learn the continuous motion dynamics and ensure temporally consistent results [8]. Furthermore, many of these approaches require additional preprocessing steps for key-frame identification, introducing practical challenges in clinical settings. Recent work, notably EchoTracker [9], has advanced the field by introducing a landmark tracking framework capable of analyzing complete cardiac cycles. However, its reliance on large-scale, densely annotated video sequences poses a significant barrier to clinical adoption, as such comprehensive annotations are rarely available in practical clinical scenarios.

Despite significant advances, current state-of-the-art methods still face three critical limitations that hinder their comprehensive clinical adoption: 1) Most approaches rely on pre-defined key frames, requiring additional preprocessing steps for frame identification during inference. This dependency prevents true end-to-end learning and introduces both computational overhead and potential inconsistency in frame selection. 2) The sparse nature of key-frame annotations in typical clinical workflows poses a fundamental challenge. With no annotations available for intermediate frames, existing methods are constrained to key-frame supervision, which cannot yield accurate and consistent landmark predictions across time, leading to poor temporal consistency and inadequate understanding of dynamic cardiac motion patterns. 3) While video-based approaches aim to leverage temporal information, the computational complexity of modeling dense spatial-temporal relationships becomes prohibitive in practice, substantially impairing the effective modeling of both inter- and intra-frame relationships in clinical applications.

To address these challenges, we propose SemiEchoTracker, a novel semi-supervised framework enabling end-to-end automated landmark tracking throughout entire echocardiography sequences while requiring supervision only on key frames. Our framework introduces three key innovations: First, we develop an innovative co-training mechanism that integrates a spatial detector with a temporal tracker. The detector provides initialization for each frame, while the tracker generates temporally consistent predictions for intermediate frames to guide the detector. This synergistic interaction maintains mutual consistency between the two predictors and enables effective co-training without additional annotations. Second, we introduce a guided DINOv2 pretraining strategy that leverages abundant unlabeled frames in sparsely annotated videos. Rather than conventional random masking, this approach employs pre-defined landmark proposals, enabling more effective model pretraining specifically tailored to echocardiographic features. Third, we present a Perception-Aware Spatial-Temporal (PAST) attention module that efficiently transforms single-frame spatial features from the pretrained DINOv2 model into temporal-aware representations while maintaining computational efficiency. Upon completion of training, SemiEchoTracker delivers a robust, temporal-aware detector that explicitly learns the motion dynamics and capable of performing comprehensive landmark tracking across entire video sequences without needing predefined key frames. Our contributions can be summarized as follows:

1) A novel semi-supervised framework for echocardiography landmark tracking, which integrates spatial detection with temporal tracking to maintain consistency and enhance performance on both key and intermediate frames without additional annotations.
2) A guided DINOv2 pretraining strategy that effectively leverages large-scale unlabeled echocardiography data to build a robust spatial feature extractor, particularly for echocardiography, maximizing the utility of sparsely annotated datasets.
3) A perception-aware spatial-temporal (PAST) attention module that refines extracted spatial features into spatial- and temporal-aware representations while maintaining computational efficiency.
4) Excellent performance of our approach in both detection and tracking tasks, as demonstrated through comprehensive experiments on three datasets with different cardiac views.

The remainder of this paper is organized as follows. Section II presents the related work. Section III describes the proposed SemiEchoTracker framework in detail. Section IV presents the experimental setup. Experimental results on three datasets as well as the ablation study are provided in Section V. Finally, Section VII concludes the paper.

II. Related Work

A. Landmark Detection in Medical Images

The advent of deep learning [10], [11] has transformed data-driven landmark detection, with existing approaches broadly categorized into three methodological streams: regression-based, heatmap-based, and graph-based methods. Regression-based approaches [12], [13] employ a feature extractor coupled with a projector for direct coordinate regression. While computationally efficient, these encoder-projector architectures often struggle with precise localization, primarily due to the inherent challenge of mapping high-dimensional image features directly to coordinate values. Heatmap-based methods [14], [15], [16], [17] address this limitation by reformulating landmark detection as a dense prediction task, preserving spatial information through probability distribution maps. However, these approaches face challenges in accurately modeling local representations for complex anatomical structures. Recent work such as CeLDA [15] has advanced this approach by enhancing localization accuracy through the computation of similarities between image features and landmark prototypes extracted from local representations. Graph-based methods inherently incorporate structural constraints through graph neural networks (GNNs), addressing a crucial aspect of anatomical landmark detection that requires explicit architectural modifications in other approaches. DAG [18] combines CNN feature extraction with dual GNNs, employing mean shape initialization followed by iterative refinement through

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global and local graph networks. However, its reliance on mean shape initialization potentially limits generalizability across diverse medical datasets. EchoGraph [6] and EchoSTGNet [7] propose to transform CNN-extracted features into an implicit graph structure without any pre-defined shape initialization, enabling direct coordinate regression without explicit coordinate modeling or initialization.

B. Landmark Tracking in Echocardiography Video

Landmark tracking in video data, particularly in echocardiography, presents unique challenges beyond static landmark detection [19], [20], [21]. The standard clinical workflow in single-view 2D echocardiography typically provides annotations only for key frames, leaving intermediate frames unlabeled. This sparse annotation pattern has constrained most existing methods [5], [6], [7], [22] to key-frame detection only, limiting their ability to learn the motion dynamics. While some approaches [6], [7] claim capabilities in intermediate frame detection, they often suffer from temporal inconsistency and lack comprehensive validation across complete cardiac cycles with explicit modeling. Recent work by EchoTracker [9] has made significant progress by introducing a framework capable of analyzing entire cardiac cycles. However, its effectiveness relies heavily on a large-scale dataset with dense frame-wise annotations across time and numerous landmarks, a requirement that proves impractical in clinical settings where such comprehensive annotations are both resource-intensive and prohibitively expensive to obtain.

C. Temporal Dependency in Echocardiography

Echocardiography sequences inherently exhibit strong temporal continuity, characterized by gradual and continuous changes in anatomical structures across consecutive frames. Leveraging this temporal information has proven crucial for enhancing model performance. Current approaches to temporal modeling in echocardiography primarily follow two methodological paradigms. The first approach [6] utilizes 3D CNN architectures to encode image sequences, incorporating temporal information through 3D convolution operations. While conceptually straightforward, these methods suffer from high computational costs and their feature interactions are constrained to local regions defined by kernel sizes, which may restrict the model's ability to capture meaningful relationships across both spatial and temporal dimensions. The second approach employs specialized recurrent modules such as Long Short-Term Memory (LSTM) [23] or Gated Recurrent Units (GRU) [24] to explicitly model temporal relationships. For instance, Echo-STGNet [7] employs GRU modules to establish frame-to-frame temporal connections. However, these recurrent architectures demonstrate inherent limitations in capturing comprehensive temporal patterns. Their sequential processing nature and limited memory mechanisms often struggle to effectively model long-range dependencies, both between frames and within spatial regions, potentially missing critical patterns in cardiac motion sequences. These limitations highlight the need for more sophisticated temporal modeling approaches in echocardiography analysis. An ideal solution must simultaneously address multiple challenges: maintaining computational efficiency for clinical video sequence processing, effectively capturing both short-range and long-range spatial-temporal dependencies across the cardiac cycle, and preserving the fine-grained spatial information that is crucial for accurate landmark localization. These requirements motivate the development of more advanced temporal modeling strategies that can better serve echocardiography tracking.

III. Methodology

A. Problem Formulation

Given an echocardiography video dataset $D = \{V_1, V_2, \cdots, V_M\}$, where video sequence $V_i = \{I_1, I_2, \cdots, I_t, \cdots, I_T\}$ consists of T frames, with I_t representing the t-th frame. For clinical relevance, we consider the end-diastolic (ED) and end-systolic (ES) frames as the first and last frames of each sequence, respectively. The available annotations are denoted as $Y = \{Y_1, Y_2, \cdots, Y_M\}$, where $Y_i = \{\mathbf{L}_{ED}, \mathbf{L}_{ES}\}$ represents the annotations for the paired ED and ES frames in the i-th video sequence. Here, $\mathbf{L}_{ED} = \{\mathbf{L}^1_{ED}, \mathbf{L}^2_{ED}, \cdots, \mathbf{L}^N_{ED}\}$ and $\mathbf{L}_{ES} = \{\mathbf{L}^1_{ES}, \mathbf{L}^2_{ES}, \cdots, \mathbf{L}^N_{ES}\}$ denote the coordinates of N landmarks on the ED and ES frames, respectively. Current methods, constrained by such sparse annotations, fail to learn the motion dynamics across video sequences. We address this limitation through our proposed SemiEchoTracker, a semi-supervised landmark tracking framework with key frames supervision only. The overall framework is illustrated in Fig. 1. In Section III-B, we first present a guided DINOv2 pretraining strategy that enables the extraction of echocardiography-specific fine-grained spatial features. In Section III-C, we then propose our co-training mechanism that leverages both spatial detector and temporal tracker to complementarily enforce the consistency of the intermediate frames across the entire video sequence. To enhance the spatial-temporal relationships of the extracted features, we further introduce a perception-aware spatial-temporal (PAST) attention module in Section III-D for effectively capturing the inter- and intra-frame relationships while maintaining computational efficiency.

B. Guided DINOv2 Pretraining

Clinical echocardiography workflows typically involve sparse temporal annotation, where only key frames (ED, ES) are labeled, resulting in a substantial volume of unlabeled intermediate frames. While previous methods have largely overlooked this unlabeled data, we recognize its potential value in capturing essential echocardiographic characteristics. To effectively leverage these abundant unlabeled frames, we incorporate DINOv2 [25], a state-of-the-art self-supervised learning framework, for pretraining our spatial feature extractor. However, echocardiographic images present unique challenges that necessitate modifications to conventional pretraining approaches. The original DINOv2 employs random patch masking to train the model for image reconstruction, but this approach proves to be suboptimal for echocardiography. Specifically, echocardiographic images contain substantial background regions outside the probe's field

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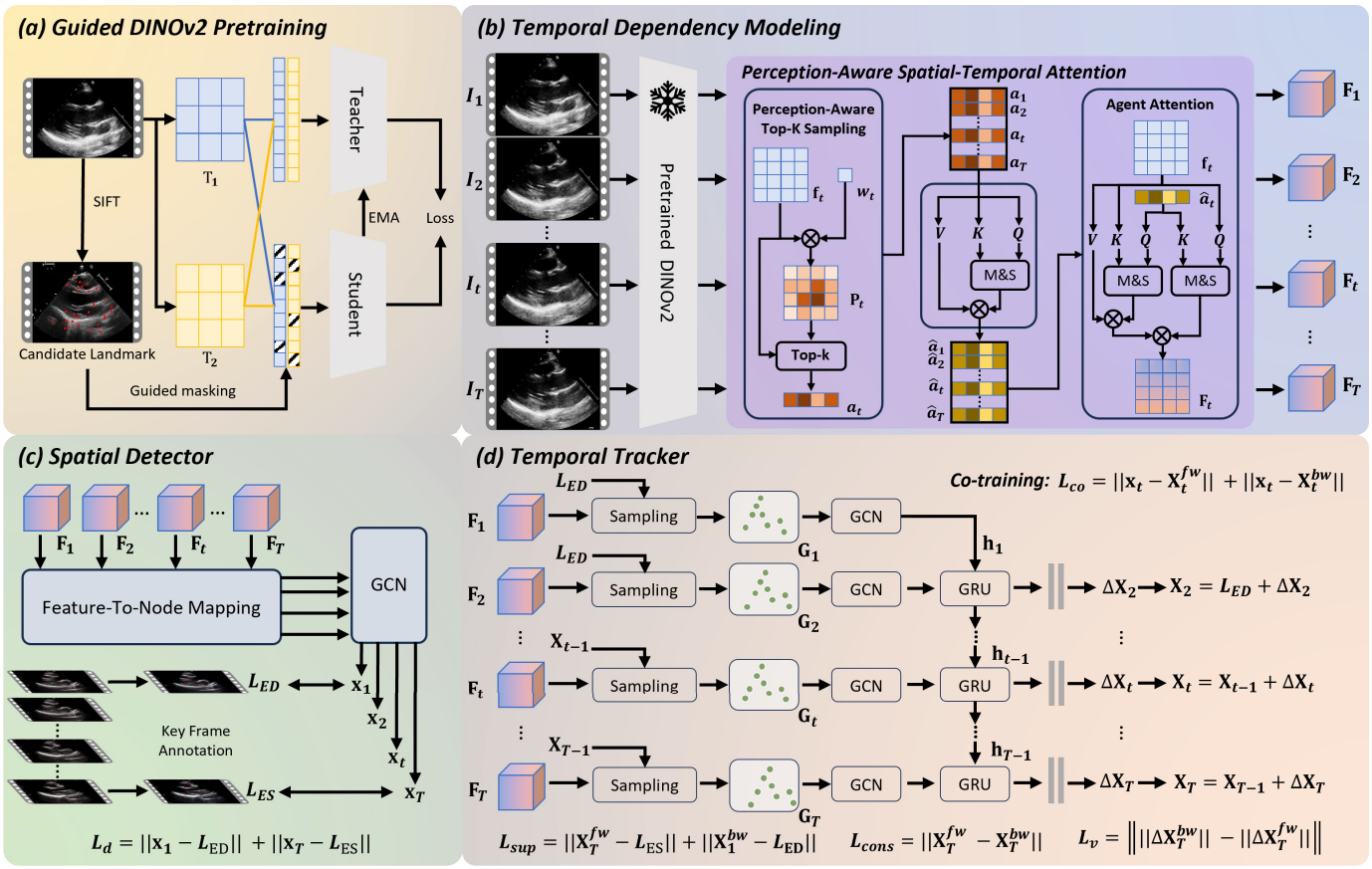


Fig. 1. Overview of our proposed framework. It begins with a pretraining with the Guided DINOv2 strategy (a). The pretrained encoder is then frozen to extract spatial features, which are subsequently processed by a PAST module (b) to capture spatio-temporal relationships. Finally, the spatial detector (c) and the temporal tracker (d) are co-trained on the intermediate frames, supervised only by the ground-truth annotations of ED/ES frames. Here, M&S represents Multiplication and Softmax operations, respectively.

of view where masking and reconstruction provide minimal learning value and may potentially hinder model training. To address this limitation, we propose a landmark-guided masking strategy to replace original random masking, enabling the model to focus on learning features from clinically relevant cardiac regions.

Formally, given an input image I_t, we employ a pre-defined landmark proposal model to generate candidate landmarks l:

$$l = \mathcal{M}(I_t), \tag{1}$$

where $\mathcal{M}$ denotes the landmark proposal model, i.e. SIFT [26] or SuperPoint [27]. These proposals l are then mapped to their corresponding patches to guide the masking process, replacing DINOv2's random masking strategy and making the self-supervised learning focus on regions of cardiac significance.

C. Semi-Supervised Co-Training Mechanism

Our semi-supervised co-training mechanism leverages the complementary strengths of spatial detection and temporal tracking to improve landmark localization on unlabeled intermediate frames. This approach enables mutual enhancement between the detector and the tracker while maintaining consistency across the temporal sequence, as shown in Fig. 1(b,c,d).

Given an input video sequence $\mathbf{V}_i = \{I_1, I_2, \cdots, I_t, \cdots, I_T\}$ where T denotes the sequence length, we first extract frame-level spatial features through our pretrained backbone network $\mathcal{E}$:

$$\mathbf{f}_t, \mathbf{w}_t = \mathcal{E}(I_t), \quad t \in \{1, \cdots, T\} \tag{2}$$

where $\mathbf{f}_t \in \mathbb{R}^{C\times h\times w}$ represents the spatial features of frame t, with C, h, and w denoting the channel dimension, height, and width, respectively. $\mathbf{w}_t \in \mathbb{R}^{1\times C}$ represents the class token of the frame t. To capture temporal dependency, we process these spatial features through our PAST attention module $\mathcal{P}$, which will be detailed in Sec. III-D:

$$\begin{aligned}\mathbf{F} &= \{\mathbf{F}_1, \mathbf{F}_2, \cdots, \mathbf{F}_t, \cdots, \mathbf{F}_T\} \\ &= \mathcal{P}(\{[\mathbf{f}_1, \mathbf{w}_1], [\mathbf{f}_2, \mathbf{w}_2] \cdots, [\mathbf{f}_t, \mathbf{w}_t], \cdots, [\mathbf{f}_T, \mathbf{w}_T]\})\end{aligned} \tag{3}$$

where $\mathbf{F}_t \in \mathbb{R}^{C\times h\times w}$ represents the enhanced spatial-temporal features incorporating both intra- and inter-frame relationships for the t-th frame.

1) Spatial Detector: The spatial detector $\mathcal{D}$ operates on the enhanced features $\mathbf{F}$ to predict frame-wise landmark coordinates as shown in Fig. 1(c). Specifically, the detector first transforms these dense features $\mathbf{F}$ into a graph representation through a feature-to-node (FTN) module, implemented as a series of convolutional layers with the output channels matching the number of landmarks N. For the t-th frame with input feature $\mathbf{F}_t$, the FTN module generates encoded features:

$$\mathbf{g}_t = \mathbf{FTN}(\mathbf{F}_t), t \in \{1, 2, \cdots, T\} \tag{4}$$

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where $\mathbf{g}_t \in \mathbb{R}^{N\times\frac{h}{4}\times\frac{w}{4}}$ represents the encoded feature map, with N denoting the number of landmark nodes, which can be further reshaped into a graph structure $\in \mathbb{R}^{N\times\frac{hw}{16}}$ with $\frac{hw}{16}$ representing the node feature dimension.

Landmark coordinates are predicted through graph convolution operations followed by coordinate regression with an MLP as:

$$\mathbf{x}_t = \mathbf{MLP}(\mathbf{GCN}(\mathbf{g}_t)) \tag{5}$$

where $\mathbf{x}_t$ represents the predicted coordinates for frame t. The detector is supervised using key-frame ground truth:

$$\mathcal{L}_d = \|\mathbf{x}_1 - \mathbf{L}_{ED}\| + \|\mathbf{x}_T - \mathbf{L}_{ES}\| \tag{6}$$

2) Temporal Tracker: The temporal tracker enhances landmark predictions by explicitly modeling the temporal consistency of landmarks across time as shown in Fig. 1(d). Starting with the first frame, we construct a graph structure $\mathbf{G}_1 \in \mathbb{R}^{N\times[C+2(N-1)]}$ using features $\mathbf{F_1}$ and the initial ground-truth annotation $\mathbf{L}_{ED}$. Unlike the detector's implicit graph representation, the tracker incorporates explicit anatomical information through node features $\mathbf{n} \in \mathbb{R}^{N\times C}$ and shape features $\mathbf{s} \in \mathbb{R}^{2(N-1)}$. Node features $\mathbf{n}$ are extracted by grid sampling from $\mathbf{F}_1$ at the initialized landmark coordinates: $\mathbf{n} = \mathbf{grid_sample}(\mathbf{L}_{ED}, \mathbf{F}_1)$. To capture global anatomical structure, we compute shape features as inter-landmark displacement vectors: $\mathbf{s} = \{\mathbf{L}^p_{ED} - \mathbf{L}^q_{ED}\}_{p\neq q}$. The final graph features $\mathbf{G}_1$ are formed by concatenating $\mathbf{n}$ and $\mathbf{s}$, which are processed through GCNs to obtain updated features $\mathbf{h}_1 \in \mathbb{R}^{N\times C}$, serving as the initial GRU state.

For subsequent frames I_t, we maintain the previous frame's landmark positions $\mathbf{X}_{t-1}$ ($\mathbf{L}_{ED}$ for the 2_{nd} frame) while constructing the current frame's graph using corresponding features $\mathbf{F}_t$. Specifically, at frame t, we construct graph $\mathbf{G}_t$ using features from $\mathbf{X}_{t-1}$ and $\mathbf{F}_t$. The temporal state is updated as:

$$\mathbf{H}_t, \mathbf{h}_t = \mathrm{GRU}(\mathrm{GCN}(\mathbf{G}_t), \mathbf{h}_{t-1}) \tag{7}$$

where the updated output state $\mathbf{H}_t$ is used to predict position offsets through:

$$\Delta\mathbf{X}_t = \mathrm{MLP}(\mathbf{H}_t) \quad \mathbf{X}_t = \mathbf{X}_{t-1} + \Delta\mathbf{X}_t \tag{8}$$

where $\Delta\mathbf{X}_t$ represents the predicted position offset for the t-th frame based on the initial landmark positions $\mathbf{X}_{t-1}$. The tracker's prediction $\mathbf{X}_t$ of frame t is then updated.

The tracker operates bidirectionally, processing the sequence in both forward (1 to T) and backward (T to 1) manners, producing corresponding offset predictions ($\Delta\mathbf{X}^{fw}_t$, $\Delta\mathbf{X}^{bw}_t$) and landmark predictions ($\mathbf{X}^{fw}_t$, $\mathbf{X}^{bw}_t$). For the final tracked results, we have supervisions of ground-truth landmarks at key frames as:

$$\mathcal{L}_{sup} = \|\mathbf{X}^{fw}_T - \mathbf{L}_{ES}\| + \|\mathbf{X}^{bw}_1 - \mathbf{L}_{ED}\| \tag{9}$$

To ensure temporal coherence, we enforce consistency between forward and backward predictions through:

$$\mathcal{L}_{cons} = \|\mathbf{X}^{fw}_t - \mathbf{X}^{bw}_t\|, \quad t \in \{2, 3, \cdots, T-1\} \tag{10}$$

Additionally, we maintain motion consistency by constraining velocity predictions for each time step between the forward and backward processes:

$$\mathcal{L}_v = \left\| \|\Delta\mathbf{X}^{fw}_t\| - \|\Delta\mathbf{X}^{bw}_t\| \right\|, \quad t \in \{2, 3, \cdots, T-1\} \tag{11}$$

3) Spatial-Temporal Co-Training: The spatial detector and temporal tracker exhibit complementary characteristics in landmark localization: While the detector can generate frame-wise predictions without preprocessing, but lacks temporal context, the tracker excels at maintaining temporal continuity and consistency across the sequence. We leverage this complementarity through a novel co-training strategy that utilizes the tracker's temporal coherence to enhance the detector's spatial predictions on intermediate frames.

Specifically, the co-training mechanism enforces consistency between the detector's frame-wise predictions and the tracker's temporally-aware predictions through:

$$\mathcal{L}_{co} = \|\mathbf{X}^{fw}_t - \mathbf{x}_t\| + \|\mathbf{X}^{bw}_t - \mathbf{x}_t\|, \quad t \in \{2, 3, \cdots, T-1\} \tag{12}$$

where $\mathbf{X}^f_t$ and $\mathbf{X}^b_t$ represent the forward and backward predictions of the tracker, respectively, and $\mathbf{x}_t$ denotes the detector's predictions.

Our SemiEchoTracker is optimized through a weighted combination of all loss components:

$$\mathcal{L} = \mathcal{L}_d + \lambda_1\mathcal{L}_{sup} + \lambda_2\mathcal{L}_{cons} + \lambda_3\mathcal{L}_v + \lambda_4\mathcal{L}_{co} \tag{13}$$

where $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are the balancing coefficients used to regulate the contribution of each loss.

D. Perception-Aware Spatial-Temporal Attention

As discussed in Sec. II, existing temporal modeling approaches are either inherently constricted with the local receptive field of 3D convolution operations, or limited by temporal state modules like GRU/LSTM with sparse representations, which are insufficient for dense prediction tasks (that require comprehensive spatial-temporal feature understanding). These limitations motivate us to introduce the Perception-Aware Spatial-Temporal (PAST) attention mechanism as illustrated in Fig. 1(b).

The PAST attention $\mathcal{P}$ is driven by the strong pretrained encoder to guide feature selection and attention computation. Given the class token $\mathbf{w}_t$ of the t-th frame, we first compute the perception matrix $\mathbf{P}_t$ between the class token and the image feature vector $\mathbf{f}'_t \in \mathbb{R}^{hw\times C}$ which is reshaped from $\mathbf{f}_t \in \mathbb{R}^{C\times h\times w}$, to achieve a perception awareness:

$$\mathbf{P}_t = \mathrm{softmax}\left(\frac{\mathbf{f}'_t\mathbf{w}^\top_t}{\|\mathbf{f}'_t\|\|\mathbf{w}_t\|}\right) \tag{14}$$

where softmax$(\cdot)$ denotes the softmax operation. As demonstrated in [25], the pretrained DINOv2 can generate high-quality perception matrices $\mathbf{P}_t$ where higher values indicate more semantically relevant spatial features. We leverage this property by implementing a **Top-K** selection operation to extract the k most relevant tokens $\mathbf{a}_t \in \mathbb{R}^{k\times C}$ from the spatial features by $\mathbf{a}_t = \mathbf{Top\text{-}K}(\mathbf{P}_t, k, \mathbf{f}_t)$. These selected tokens are then concatenated across all frames to form $\mathbf{a} \in \mathbb{R}^{Tk\times C}$: $\mathbf{a} = \mathrm{concat}(\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_t, \cdots, \mathbf{a}_T)$.

Computing spatial-temporal attention directly on the full feature set $\mathbf{f}$ would be computationally prohibitive, as it

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involves attention computation over $\mathbb{R}^{Thw\times C}$ dimensions. Moreover, ultrasound data typically contains background regions that do not warrant comprehensive attention computation. Instead, our approach utilizes $\mathbf{a} \in \mathbb{R}^{Tk\times C}$, which is guided by the pretrained DINOv2, represents an aggregation of the most informative spatial features across all frames, with $k \ll h \times w$. This enables efficient spatial-temporal attention computation:

$$\hat{\mathbf{a}} = \text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{MultiHead}(\mathbf{a}, \mathbf{a}, \mathbf{a}) \tag{15}$$

The resulting temporal-spatial-aware tokens $\hat{\mathbf{a}} \in \mathbb{R}^{Tk\times C}$ are then split into frame-wise representations: $\hat{\mathbf{a}} = \{\hat{\mathbf{a}}_1, \hat{\mathbf{a}}_2, \cdots, \hat{\mathbf{a}}_t, \cdots, \hat{\mathbf{a}}_T\}$. Subsequently, for time step at t, we perform agent attention [28] between spatial features and temporal-spatial-aware tokens as below:

$$\mathbf{A}_t = \text{softmax}\left(\frac{\hat{\mathbf{a}}_t\mathbf{W}^Q(\mathbf{f}_t\mathbf{W}^K)^\top}{\sqrt{C}}\right)\mathbf{f}_t\mathbf{W}^V \tag{16}$$

$$\mathbf{F}'_t = \text{softmax}\left(\frac{\mathbf{f}_t\mathbf{W}^Q(\hat{\mathbf{a}}_t\mathbf{W}^K)^\top}{\sqrt{C}}\right)\mathbf{A}_t \tag{17}$$

where $\mathbf{F}'_t \in \mathbb{R}^{hw\times C}$ represents the enhanced spatial features with both local spatial information and global temporal context, which can be reshaped into $\mathbf{F}_t \in \mathbb{R}^{C\times h\times w}$.

IV. Experiments Setup

A. Datasets

We evaluate our method on three echocardiography datasets representing different cardiac anatomical views to demonstrate the robustness and generalizability of our approach.

1) Pretraining Dataset: A large-scale dataset was curated for pre-training, which was collected from the Second Affiliated Hospital of Wenzhou Medical University and Wenzhou People's Hospital, comprising 25,628 video sequences and approximately 2.5 million frames across two standard views: Apical Four-Chamber (A4C) and Parasternal Long-Axis (PLAX).

2) PLAX Dataset: The parasternal long axis (PLAX) dataset comprises 260 echocardiography video sequences acquired from 260 patients at The Second Affiliated Hospital of Wenzhou Medical University and Wenzhou People's Hospital, using B-mode transthoracic echocardiography. Each sequence captures at least one complete cardiac cycle. The dataset is randomly partitioned into training (200), validation (30), and testing (30) sets. Following standard clinical protocols, the training set contains annotations exclusively for end-diastolic (ED) and end-systolic (ES) frames, while the testing set includes comprehensive frame-by-frame annotations spanning the complete cardiac cycle between ED and ES frames, enabling evaluation of temporal tracking performance. The dataset includes 11 anatomical landmarks annotated to measure critical cardiac dimensions: left atrial anteroposterior diameter, aortic sinus diameter, left ventricular outflow tract diameter, right ventricular anteroposterior diameter, interventricular septal thickness, left ventricular posterior wall thickness, and left ventricular internal diameter.

3) A4C Dataset: The apical four-chamber (A4C) dataset consists of 320 echocardiography video sequences collected from 320 patients at The Second Affiliated Hospital of Wenzhou Medical University and Wenzhou People's Hospital. Following identical acquisition and annotation protocols as the PLAX dataset, the data is systematically divided into training (240), validation (30), and testing (50) sets. The training set similarly contains sparse annotations limited to ED and ES frames, while the testing set incorporates dense frame-by-frame annotations to enable rigorous evaluation of temporal consistency and tracking accuracy. The dataset includes three anatomical landmarks: the cardiac apex and the anterior and posterior leaflets of the mitral valve.

4) EchoNet-Dynamic Dataset: The EchoNet-Dynamic dataset is a large-scale echocardiography dataset comprising 10,030 samples, with each sample containing only 4CH-view echocardiography cine. We randomly sampled 560 samples from this dataset, which is split into training (400), validation (80), and testing (80) sets, respectively. The intermediate frames of validation and testing sets are manually annotated by two expert radiologists to report the performance.

B. Evaluation Metrics

We evaluate landmark performance using two standard metrics in medical image analysis: Mean Radial Error (MRE) and Success Detection Rate (SDR). The MRE quantifies the average Euclidean distance between predicted landmarks and their corresponding ground-truth positions:

$$\text{MRE} = \frac{1}{N}\sum_{j=1}^{N}\|\hat{\mathbf{y}}_j - \mathbf{y}_j\| \tag{18}$$

where $\hat{\mathbf{y}}_j$ and $\mathbf{y}_j$ denote the predicted and ground-truth positions of the j-th landmark, respectively, and N represents the total number of landmarks.

The Success Detection Rate (SDR) assesses prediction accuracy by measuring the proportion of landmarks correctly localized within a specified threshold σ:

$$\text{SDR} = \frac{1}{N}\sum_{j=1}^{N}\mathbb{I}(\hat{\mathbf{y}}_j - \mathbf{y}_j < \sigma) \tag{19}$$

where $\mathbb{I}(\cdot)$ represents the indicator function.

C. Implementation Details

Our framework implementation comprises two primary stages: pretraining and co-training. For the pretraining stage, we employ a Vision Transformer (ViT-S) architecture with a patch size of 14 as our backbone. The model is trained on all datasets strictly under the default setting of DINOv2 [25], except its masking strategy is replaced with the guided masking as introduced in Sec. III-B. We set the learning rate and weight decay to 1×10^{-4} and 1×10^{-5}, and standardize all input images to 224×224 pixels. The pretraining is conducted on a distributed system with 8 NVIDIA A100 GPUs, utilizing a total batch size of 8×196. In the co-training stage, we utilized the pretrained ViT-S backbone as the feature extractor (frozen) and trained the subsequent detector

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TABLE I
EXPERIMENTAL RESULTS OF DIFFERENT METHODS WITH TRACKING PERFORMANCE ON PLAX DATASET

Method	Detection				Tracking			
	MRE ↓ (mm, std)	SDR ↑ (%)			MRE ↓ (mm, std)	SDR ↑ (%)		
		2mm	3mm	4mm		2mm	3mm	4mm
UNet [29]	4.12 (4.61)	32.58	52.39	66.30	4.95 (4.73)	28.33	44.39	54.85
HRNet [30]	4.16 (4.38)	33.03	49.61	62.27	5.16 (5.70)	28.79	41.52	51.97
HYATT-Net [16]	3.01 (2.98)	40.17	66.71	79.43	3.42 (4.31)	39.10	50.45	75.18
CeLDA [15]	2.97 (3.16)	42.27	68.61	82.39	3.19 (3.37)	38.09	59.39	76.52
U-LanD [5]	3.84 (0.84)	27.27	50.76	70.45	3.97 (0.90)	25.55	47.55	67.33
EchoGLAD [22]	2.93 (3.42)	42.39	68.27	82.00	3.30 (4.27)	38.03	59.85	75.61
EchoGraphs [6]	2.98 (0.30)	41.88	73.05	87.66	3.04 (0.51)	29.15	56.00	77.45
EchoSTGNet [7]	2.58 (1.21)	46.13	66.06	87.81	2.99 (1.60)	36.75	65.01	84.68
Ours	**2.21 (0.27)**	**48.79**	**75.91**	**91.36**	**2.23 (0.37)**	**49.30**	**75.82**	**90.97**

TABLE II
EXPERIMENTAL RESULTS OF DIFFERENT METHODS WITH TRACKING PERFORMANCE ON A4C DATASET

Method	Detection				Tracking			
	MRE ↓ (mm, std)	SDR ↑ (%)			MRE ↓ (mm, std)	SDR ↑ (%)		
		2mm	3mm	4mm		2mm	3mm	4mm
UNet [29]	2.80 (2.51)	46.43	66.23	81.66	2.93 (2.76)	44.67	61.07	80.74
HRNet [30]	2.50 (1.70)	48.17	71.56	84.22	2.85 (2.10)	47.89	67.50	81.67
HYATT-Net [16]	2.41 (1.10)	44.37	70.01	87.78	2.52 (3.13)	46.04	75.93	85.37
CeLDA [15]	2.32 (2.27)	51.54	74.72	85.79	2.45 (2.63)	48.46	74.86	86.52
U-LanD [5]	2.77 (1.78)	46.07	62.55	81.42	2.96 (1.75)	41.69	60.66	82.51
EchoGLAD [22]	2.39 (1.54)	48.33	73.33	84.11	2.59 (1.51)	45.48	69.26	84.89
EchoGraphs [6]	2.19 (1.43)	50.56	79.44	92.22	2.38 (1.44)	47.93	74.22	87.70
EchoSTGNet [7]	2.35 (1.34)	45.37	71.26	90.78	2.49 (1.44)	35.00	72.22	89.44
Ours	**2.13 (0.45)**	**52.74**	**81.41**	**94.07**	**2.14 (0.64)**	**49.44**	**86.67**	**94.44**

TABLE III
EXPERIMENTAL RESULTS OF DIFFERENT METHODS WITH TRACKING PERFORMANCE ON ECHONET-DYNAMIC DATASET

Method	Detection				Tracking			
	MRE ↓ (mm, std)	SDR ↑ (%)			MRE ↓ (mm, std)	SDR ↑ (%)		
		2mm	3mm	4mm		2mm	3mm	4mm
CeLDA [15]	2.74 (2.89)	45.82	71.36	81.94	3.02 (3.12)	40.73	63.85	75.26
EchoGraphs [6]	2.45 (1.85)	48.91	74.27	83.52	2.68 (2.01)	45.18	68.94	79.37
EchoSTGNet [7]	2.31 (1.64)	50.25	75.83	86.04	2.47 (1.78)	48.61	71.52	81.73
Ours	**2.13 (1.07)**	**54.17**	**77.50**	**89.17**	**2.15 (1.37)**	**54.00**	**76.33**	**88.17**

and tracker for 300 epochs. We set the learning rate to 1×10^{-3} with Adam optimizer and CosineAnnealing learning schedule on a single NVIDIA V100 GPU with a batch size of 4. The parameter k used in **Top-K** selection with PAST is set to 32. We have also made the source code publicly available at the following GitHub repository: https://github.com/Fitz-Fitz/SemiEchoTracker.

All experiments follow a rigorous evaluation protocol: Models are trained on the training set, validated on the validation set for ablation and hyper-parameter tuning, and final results are reported on the held-out test set for all dataset. During inference, we employ only the distilled detector for landmark detection across all frames, ensuring efficient deployment while maintaining performance.

V. EXPERIMENTS RESULTS

A. Comparing Methods

We compare our proposed SemiEchoTracker with leading methods in echocardiographic landmark detection, including an uncertainty-based method U-LanD [5], one static graph-based detection method EchoGLAD [22], and two video-based temporal-aware detection methods such as EchoGraph [6] with 3D CNN as temporal modeling and EchoSTGNet [7] with GRU as temporal modeling. Furthermore, we extend our comparison to include SOTA landmark detection methods from other domains: one cephalometric landmark detection approach CeLDA [15], one unified X-Ray landmark detection method HYATT-Net [16], and also two classical heatmap-based approaches UNet [29] and HRNet [30].

B. Landmark Evaluation

We evaluate our method's performance on both detection and tracking tasks across two standard echocardiographic views: Parasternal Long-Axis (PLAX) and Apical Four-Chamber (A4C). For a comprehensive evaluation, we employ two metrics: Mean Radial Error (MRE) in millimeters with standard deviation, and Successful Detection Rate (SDR) at different precision thresholds (2mm, 3mm, and 4mm). We evaluate both **Detection** performance (on key frames) and **Tracking** performance (across entire sequence) using these metrics on both datasets.

1) Performance on PLAX Dataset: Table I presents comprehensive evaluation results on the PLAX dataset. A notable observation is the significant performance gap between detection and tracking tasks in existing methods, highlighting the challenge of maintaining consistent performance across temporal sequences. Our method effectively addresses this limitation, demonstrating superior performance across all metrics. Specifically, in the detection task, our method achieves an MRE of 2.21mm with a remarkably low standard deviation, representing a 14.3% improvement over the previous state-of-the-art EchoSTGNet (2.58mm). The improvement is particularly pronounced in SDR metrics, where our method achieves 48.79%, 75.91%, and 91.36% at 2mm, 3mm, and 4mm thresholds respectively, demonstrating consistent advantages across varying precision requirements.

The tracking results further validate our method's effectiveness in maintaining temporal consistency. Notably, we maintain nearly identical performance between detection and tracking tasks, while other methods exhibit significant performance degradation in tracking. It is also observed from Fig. 2 that, while existing methods perform adequately on keyframe detection (i.e., the first and last frames), they struggle with intermediate frames, showing both reduced accuracy and poor temporal consistency in landmark predictions. This observation further validates the superiority of our approach in maintaining both accuracy and temporal consistency throughout the entire video sequence, not just at keyframes.

2) Performance on A4C Dataset: Table II presents evaluation results on the A4C dataset, further validating our method's effectiveness across different cardiac views. In the detection task, our method achieves an MRE of 2.13mm, demonstrating substantial improvement over existing approaches. The performance improvement is particularly evident in SDR metrics, where our method achieves superior success rates across all precision thresholds. The tracking results further show our method's robust temporal modeling capabilities.

3) Performance on EchoNet-Dynamic Dataset: As demonstrated in Table III, our method achieves superior performance in both detection and tracking tasks on the public dataset. Notably, the results exhibit consistent patterns with our

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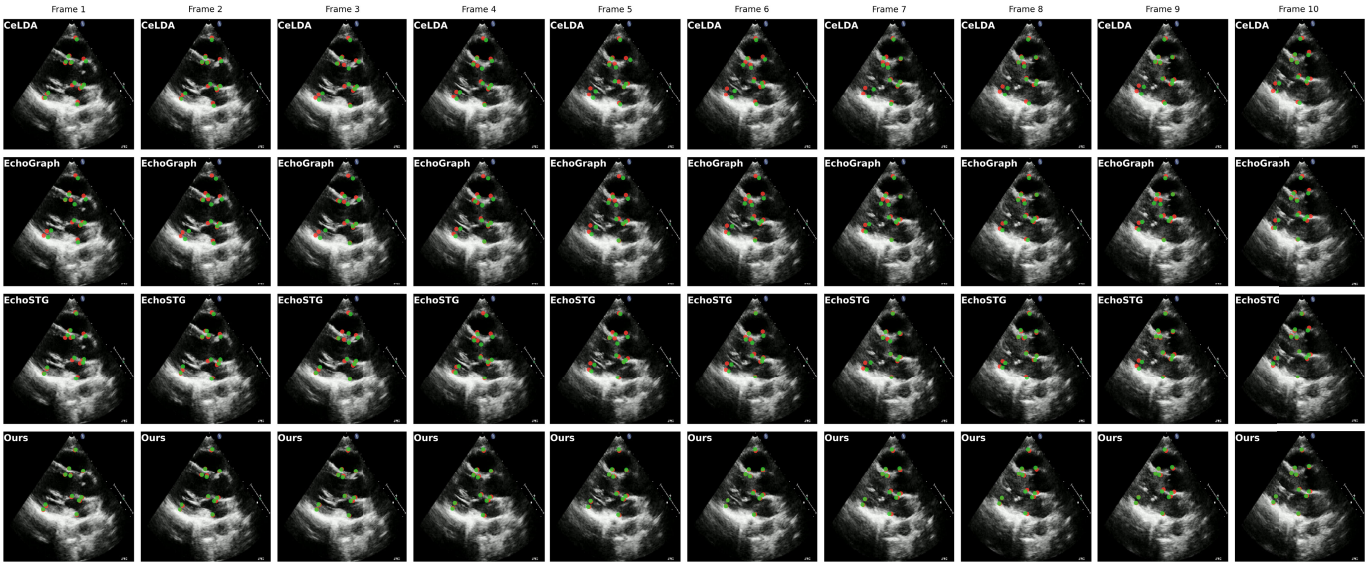


Fig. 2. Visualization results of CeLDA [15], EchoGraph [6], EchoSTGNet [7], and our methods on the PLAX dataset, where red points and green points represent the ground-truth and predicted landmarks, respectively.

TABLE IV
ABLATION STUDY OF DIFFERENT COMPONENTS IN OUR METHOD, CONDUCTED ON THE PLAX DATASET (ON VALIDATION SET)

Component	Detection		Tracking	
	MRE ↓ (mm, std)	SDR ↑ @2mm (%)	MRE ↓ (mm, std)	SDR ↑ @2mm (%)
Baseline	2.75 (1.76)	41.97	2.98 (1.81)	36.41
+ Guided Pretraining	2.64 (1.57)	43.37	2.81 (1.69)	39.36
+ Co-training	2.45 (0.92)	45.30	2.49 (0.94)	45.67
+ PAST Attention	**2.33 (0.37)**	**46.89**	**2.36 (0.67)**	**46.30**

in-house dataset evaluations, where our tracking accuracy substantially outperforms competing methods while simultaneously improving detection performance. This consistency across different datasets validates robustness and generalizability of our proposed approach, demonstrating its potential for broader clinical application beyond specific imaging protocols and patient populations of our initial dataset.

C. Ablation Studies

We conduct comprehensive ablation experiments on the PLAX dataset to evaluate the contribution of each key component in our framework. Table IV presents quantitative results across both detection and tracking tasks. Our baseline implements the original DINOv2 pretraining with an independent spatial detector, without any temporal modeling strategy, and no co-training between detector and tracker components.

1) Guided DINOv2 Pretraining: The guided DINOv2 pretraining strategy yields substantial improvement over the baseline, reducing detection MRE from 2.75mm to 2.64mm and tracking MRE from 2.98mm to 2.81mm. These results validate that our landmark-guided masking strategy can effectively focus the model's attention on clinically relevant cardiac regions during pretraining, enabling more robust feature extraction compared to conventional random masking approaches. However, the persistent performance gap between tracking and detection tasks suggests room for further improvement.

2) Co-Training Strategy: The co-training mechanism further enhances performance by exploiting the complementary strengths of spatial detection and temporal tracking. This component reduces the MRE to 2.45mm and 2.49mm for detection and tracking respectively, while significantly improving prediction stability (std of 0.92 and 0.94). The SDR metrics show substantial improvement at the 2mm threshold. These results demonstrate that our co-training strategy explicitly provides learning objectives for intermediate frames, enabling the model to better learn dynamic motion patterns, which benefit both detection and tracking tasks.

3) PAST Attention Module: The incorporation of the Perception-Aware Spatial-Temporal (PAST) attention module yields the most significant performance, achieving optimal MREs of 2.33mm and 2.36mm for detection and tracking respectively, with notably low standard deviations. The SDR metrics reach peak values of 46.89% and 46.30% at the 2mm threshold. The dramatic reduction in standard deviation demonstrates the module's effectiveness in maintaining consistent prediction quality across diverse frames and sequences, which validates that the PAST module successfully captures both inter- and intra-frame relationships.

The progressive improvement demonstrates effectiveness of our framework's components. Each addition of components contributes meaningfully to the final performance, with their combination yielding SOTA results in both detection and tracking tasks. The consistent improvement in both accuracy and stability validates our proposed strategies.

4) Choice of Self-Supervised Models: We have conducted an additional comparison of DINOv2 with other self-supervised learning methods, including MAE and DINO, with the results presented in Table V. As can observe, DINOv2 yields the best performance compared to other commonly-used

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TABLE V
ABLATION STUDY OF DIFFERENT SELF-SUPERVISED LEARNING METHODS, CONDUCTED ON THE PLAX DATASET (ON VALIDATION SET)

Method	Masking Strategy	Detection		Tracking	
		MRE ↓ (mm, std)	SDR ↑ @2mm (%)	MRE ↓ (mm, std)	SDR ↑ @2mm (%)
None	None	2.81 (1.31)	39.11	2.99 (1.14)	35.31
DINO [31]	None	2.71 (0.92)	41.10	2.88 (1.40)	39.19
MAE [32]	Random	2.59 (1.26)	43.37	2.65 (1.00)	42.41
	SuperPoint	2.52 (0.98)	43.88	2.65 (1.31)	41.11
	SIFT	2.50 (0.88)	44.78	2.59 (1.30)	41.99
DINOv2 [25]	Random	2.45 (0.39)	44.99	2.49 (0.77)	43.30
	SuperPoint	2.39 (0.92)	46.00	2.44 (1.40)	44.57
	SIFT	**2.33 (0.37)**	**46.89**	**2.36 (0.67)**	**46.30**

TABLE VI
ABLATION STUDY OF DIFFERENT TEMPORAL MODULE IN TRACKER, CONDUCTED ON THE PLAX DATASET (ON VALIDATION SET)

Component	Detection		Tracking	
	MRE ↓ (mm, std)	SDR ↑ @2mm (%)	MRE ↓ (mm, std)	SDR ↑ @2mm (%)
GAT [33]	2.35 (0.57)	45.19	2.38 (0.59)	44.79
LSTM [23]	2.39 (1.12)	44.60	2.44 (1.00)	43.67
GRU [24]	**2.33 (0.37)**	**46.89**	**2.36 (0.67)**	**46.30**

self-supervised learning methods. This further justifies the choice of DINOv2 as our pretraining model.

5) Choice of Landmark Proposal Networks: We have conducted additional experiments with full components to compare the performance of different landmark proposal networks during pretraining, including traditional computation-based method (SIFT [26]) and learning-based method (SuperPoint [27]). The results are presented in Table V. We can observe that both SIFT- and SuperPoint-based pretrained models demonstrate superior performance compared to the original random masking with different pretraining. While the performance gap between SIFT and SuperPoint is relatively modest, SIFT achieves slightly better results across all evaluation metrics. Based on these empirical findings, we select SIFT as a landmark proposal method for our Guided Masking Strategy in the final implementation. The rationale for employing guided masking is to direct the model's attention during self-supervised learning toward informative regions within the ultrasound plane, rather than uninformative black background areas. Importantly, the precision of landmark proposals is not critical for our pretraining strategy. Specifically, we do not require them to precisely identify specific anatomical structures. Rather, we only need them to roughly indicate regions within the ultrasound plane, which is substantially more effective than random masking.

6) Choice of Temporal Modeling Modules in Tracker: We have conducted additional experiments to compare GRU, LSTM and spatial-temporal graph attention network performance in our tracker, with the results presented in Table VI. As demonstrated, GRU achieves superior performance across evaluation metrics. This advantage likely stems from GRU's simpler gating mechanism, which proves to be more suitable, avoiding the overfitting tendency.

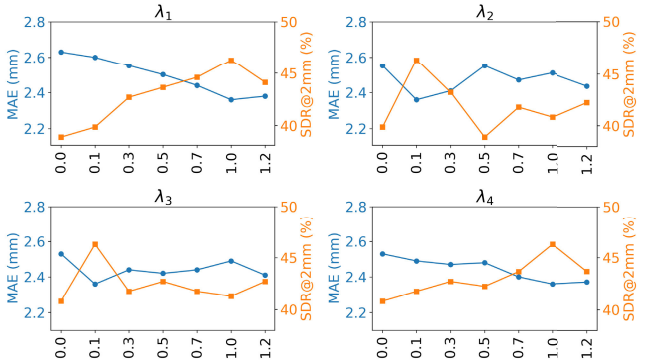


Fig. 3. Sensitivity analysis of $\lambda_1, \lambda_2, \lambda_3$, and λ_4 for each loss, conducted on PLAX dataset with tracking performance.

TABLE VII
ABLATION STUDY OF THE NUMBER OF SELECTED TOKENS k IN PAST, CONDUCTED ON THE PLAX DATASET (ON VALIDATION SET)

k	Detection		Tracking	
	MRE ↓ (mm, std)	SDR ↑ @2mm (%)	MRE ↓ (mm, std)	SDR ↑ @2mm (%)
8	2.48 (0.91)	45.11	2.53 (0.78)	44.18
16	2.38 (1.34)	45.70	2.41 (1.08)	45.32
32	2.33 (0.37)	46.89	2.36 (0.67)	46.30
64	2.35 (0.41)	45.86	2.38 (0.74)	45.11
128	2.34 (0.39)	46.18	2.36 (0.70)	45.77

D. Sensitivity Analysis of $\lambda_1, \lambda_2, \lambda_3, \lambda_4$, and k

We conduct comprehensive sensitivity analyses on the PLAX dataset to evaluate our framework's sensitivity to key hyperparameters: the loss weighting coefficients $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$ and the token selection parameter (k).

1) Loss Weighting Coefficients: We first analyze the sensitivity of loss weighting coefficients in Eq. 13, which balance different components of the overall optimization objective. Each coefficient is systematically varied from 0.0 to 1.2 while maintaining others at their optimal values. As shown in Fig. 3, the tracking performance on the PLAX dataset exhibits clear patterns with respect to these coefficients. Both extremely small and large values lead to performance degradation. Through extensive experimentation, we determine the optimal configuration to be $\lambda_1 = 1, \lambda_2 = 0.1, \lambda_3 = 0.1, \lambda_4 = 1$.

2) Token Selection Parameter: Table VII presents our analysis of the token selection parameter k in the PAST attention module, which determines the number of representative tokens selected for spatial-temporal attention computation. The results demonstrate a clear trend: both detection and tracking performance consistently improve as k increases up to 32, after which diminishing returns are observed. This pattern suggests that 32 tokens provide sufficient coverage of relevant spatial-temporal features while maintaining computational efficiency. Therefore, we adopt $k = 32$ in our final implementation.

VI. DISCUSSION AND FUTURE WORK

A. Clinical Research and Application Prospects

1) Clinical Application Validation: To demonstrate direct clinical utility, we have implemented additional experiments evaluating downstream clinical metrics. Specifically, we

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TABLE VIII
COMPARISON OF INFERENCE TIME AND COMPUTATIONAL COST

Method	Inference Time (s)	Peak GPU Memory (GB)	Parameters (MB)	EF MAE Error (%)
EchoGraph [6]	0.004	0.30	34.28	10.01
EchoSTGNet [7]	0.029	1.71	73.65	9.88
Ours	0.021	1.03	26.80	6.41

TABLE IX
USER STUDY RESULTS FOR THE PRACTICAL UTILITY OF OUR METHOD IN CLINICAL SCREENING SCENARIOS

Index	Settings	Accuracy (%)	Recall	Precision	Avg. Time (s)
D1	w/o SemiEchoTracker	85.00	0.83	0.90	77.5
	w SemiEchoTracker	92.50	0.91	0.95	41.0
D2	w/o SemiEchoTracker	82.50	0.83	0.86	69.5
	w SemiEchoTracker	95.00	0.96	0.96	48.0

assessed ejection fraction (EF) calculation accuracy using the PLAX dataset, which is a critical parameter for cardiovascular disease diagnosis, heart failure assessment, and treatment planning decisions. As reported in Table VIII, by introducing co-training and temporal information, our method achieves the minimum error compared to the gold standard.

We have also designed a simple user study to assess the clinical utility in the real-world screening workflow of our method. We collected additional 40 PLAX-view echocardiography videos that were initially diagnosed by an experienced cardiologist with 10 years of expertise in cardiac ultrasound: 23 cases were confirmed as normal after screening, and 17 cases were flagged as potentially abnormal during screening and later confirmed as abnormal with LVEF decrease after comprehensive examination. Two junior echocardiography physicians with 0.5 to 1.5 year(s) of experience were then asked to independently analyze these 40 videos in a simulated screening workflow under two conditions:

- **Condition 1 (Baseline):** Physicians reviewed the echocardiography videos alone and made screening decisions without any extra assistance.
- **Condition 2 (With SemiEchoTracker):** After a two-week washout period, two same physicians reviewed the same videos, but this time with access to our frame-by-frame landmark predictions and the derived dynamic measurement curves (e.g., wall thickness variation over time).

These results shown in Table IX demonstrating that our continuous frame-by-frame tracking approach provides tangible clinical value in realistic screening scenarios by enhancing both diagnostic confidence and efficiency.

2) Real-World Deployment Feasibility: We have conducted extensive computational performance evaluation to assess clinical deployment viability, with results presented in Table VIII. The experiments were conducted on a single NVIDIA L20 GPU with standardized input specifications: one sequence from the PLAX dataset (including 11 landmarks) containing 10 frames with a resolution of 224×224 pixels for all compared methods. We compare our methods with two video-based echo landmark detection methods: EchoGraphs [6] and EchoSTGNet [7]. EchoGraph employs a 3D CNN to extract the spatial-temporal information of the sequence with a shallow GCN for landmark detection, while EchoSTGNet utilizes a 2D CNN to extract spatial features and uses a GRU to model temporal information along the sequence. As presented in Table VIII, our method achieves a clinically viable balance between accuracy and computational efficiency with 0.021s inference time and 26.80 MB parameter count. While requiring higher peak GPU memory (1.03 GB) compared to EchoGraph (0.30 GB) and longer inference time (0.021s vs. 0.004s), this computational overhead is acceptable in clinical practice. Our method can satisfy the general hardware requirements of current clinical environments and enables real-time clinical workflow integration without disrupting routine examinations.

These comparisons with SOTA methods demonstrate that our approach achieves the optimal trade-off between performance and computational efficiency.

B. Limitations and Future Work

We have verified that our proposed method achieves superior performance in landmark detection and tracking, with excellent clinical usability. However, there are still some limitations that need to be addressed in future work.

1) Limited Clinical Verification: While our method naturally provides accurate intermediate frame-by-frame landmark tracking results, comprehensive evaluation is currently limited by the lack of established clinical gold standards for such intermediate metrics. Therefore in this work, we only focus on landmark accuracy on intermediate frames as a reliable indicator of clinical utility. Future work will explore additional clinically-relevant metrics to provide more comprehensive quantitative validation of these capabilities. Additionally, more comprehensive verification on special cases, such as patients with cardiac diseases and anatomical abnormalities, should be included in future study.

2) More Advanced Temporal Modeling With Foundation Models: For the intermediate frames, we explored how to incorporate spatial-temporal awareness during pretraining and co-training phases. However, this combination remains discrete in nature, and we envision exploring more continuous temporal modeling approaches that enable the model to naturally achieve temporal-aware perception. Future work could pursue two main directions. First, we could employ video-based methods during pretraining, allowing us to directly obtain a temporal-aware encoder for downstream tasks after pretraining. Second, we could leverage powerful video foundation models such as SAM2 [34] and SAM3 [35]. With its training on large-scale video data, SAM series has developed strong temporal reasoning capabilities. Its ability to maintain object identity across frames, even through occlusions, could theoretically be leveraged for tracking anatomical landmarks through complex cardiac motion. However, successfully adopting these models often requires extensive fine-tuning or prompt engineering, which conflicts with our goal of creating a fully automatic method that excels in the sparse-annotation settings often in clinical workflows. Addressing this challenge represents a critical direction for future research.

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VII. Conclusion

In this paper, we have presented SemiEchoTracker, a novel semi-supervised framework to enable comprehensive landmark tracking in echocardiography sequences while requiring supervision only from key frames. Our approach effectively addresses critical limitations in current methods through three key innovations: 1) a co-training mechanism that enforces mutual consistency between spatial detection and temporal tracking, enabling robust intermediate frame prediction; 2) a guided DINOv2 pretraining strategy that enhances feature extraction specifically for echocardiographic images; and 3) a perception-aware spatial-temporal (PAST) attention module that efficiently captures complex spatial-temporal dependencies while maintaining computational feasibility. Extensive validation across multiple cardiac views demonstrates the superiority of our approach, with state-of-the-art performance in both detection and tracking tasks.

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